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PS-MTL-LUCAS: A partially shared multi-task learning model for simultaneously predicting multiple soil properties

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ABSTRACT

Soil acts as a foundation for human survival and social development and soil quality has a great effect on the growth of agricultural products. Visible/near-infrared spectroscopy has been acknowledged as a rapid and non-destructive method for predicting soil properties, and multi-task learning is a preferable approach to analyze the correlation between the spectroscopy data and soil properties. However, current multi-task learning models with the soft parameter sharing structure extremely rely on the task relatedness. To tackle this limitation, we proposed PS-MTL-LUCAS, a multi-task learning with a partially shared structure in this study. An additional shared layer was utilized to learn the general informative representations and interact with each task-specific layer. The partially shared structure ensured the maximum information flow between layers, thereby boosting the prediction performance. Also, the SHapley Addictive exPlanations (SHAP) algorithm was adopted to extract the feature wavelengths of each soil property. PS-MTL-LUCAS was validated on the LUCAS topsoil dataset (2009), and the experimental result suggested that PS-MTL-LUCAS dominated state-of-the-art models by achieving the determination of coefficient at 0.945, 0.936, 0.413, 0.624, 0.837, 0.952, and 0.956 for pH, N, P, K, CEC, OC, and $CaCO_3$, respectively. In summary, this study highlighted the use of the soil spectroscopy and multi-task learning techniques in the soil property prediction task and provided a very promising approach for soil studies.

1. Introduction

Soil is the primary provider of water and nutrients. The soil fertility refers to the ability to supply nutrients in adequate quantities for plants over a sustained period of time. However, the soil fertility varies across different regions and may change with time (Lal, 2004). Therefore, rapid assessing the soil property is of great significance for making scientific decisions for soil management.

The destructive measurement like deploying physical sensors underground is a common approach of soil monitoring (Kanaya et al., 2021; Marchant, 2021; Wallor et al., 2019). The laboratory sampling measurement is also used for monitoring soil properties (Williams, 2013). However, these destructive measurements are prone to the physical damage and technical faults. Also, soil sensors are often suffered from communication issues, resulting in failed data transmission. In addition, most soil sensors have limited detection capabilities. When monitoring multiple properties, multiple soil sensors have to be deployed, leading to increasing detection costs. Therefore, a non-destructive and rapid soil monitoring method is urgently demanding.

The visible/near-infrared (Vis-NIR) spectroscopy has been adopted for assessing the soil properties in recent years. Compared with the laboratory methods and sensory measurements, The Vis-NIR spectroscopy has the advantage of rapid and non-destructive, and the spectroscopy data is usually processed by the deep learning models (Attri et al., 2023). He et al. (2023) designed an inversion model based on the decision tree algorithm to predict the nitrogen content in shallow soils. Tavakoli et al. (2023) applied the simple ratio indices and normalized difference indices to transform the raw spectral data. Various machine learning models were employed to predict multiple soil properties. Piccoli et al. (2023) proposed a deep scalable neural architecture to predict 11 soil properties and the average coefficient of determination reached around 0.75. Xia and Zhang (2022) compared the use of different satellite images for estimating the soil pH. A quantile regression forest model was implemented to determine the relationships

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between the spectral variables and the measurement. Zhang et al. (2021) adopted the hyperspectral data to assess different vegetation coverage and soil water content. A deep learning model was proposed to predict the soil water content. Lim et al. (2020) used the spectral data of the visible and near infrared region to predict the soil water content using a neural network model.

In the above related work, most researchers deployed the single-task learning models, meaning that these models can only predict a single soil property. However, soil management is not about to monitor a single soil property, but to comprehensively assess the quality of soils. Consequently, a sub branch of deep learning, namely multi-task learning (MTL), has drawn massive attention (Zhang and Yang, 2021). In general, MTL enables to learn related tasks with shared representation, thereby training a more accurate and robust learner for each task. The effectiveness of MTL models has been proved in the soil monitoring task (Gouda et al., 2021; Li et al., 2020; Padarian et al., 2019; Qi et al., 2017; Zhong et al., 2021). In these previous works, two typical MTL architectures were used, including hard parameter sharing and soft parameter sharing. In the former one, all tasks (no matter each task is related or not) share the same hidden layer and only keep a few task-specific layers for specialization in each task. Although this architecture is simple and computationally efficient, it may become less effective when one task dominates during training (Ma et al., 2018). In the latter one, each task has its own hidden layers and these sub-layers are regularized during training. The advantage of this architecture is that these task-specific layers usually have similar weights. The soft parameter sharing architecture might have troubles when the number of tasks to be learned increases because the correlation among tasks would become more complicated (Fang et al., 2022). Meanwhile, the interactions among different tasks should be carefully concerned since independent tasks might share misleading information, thereby decreasing the overall performance.

To deal with the above challenges, this study aimed at proposing a multi-task learning model to predict the value of multiple soil properties (organic carbon, cation exchange capacity, pH, calcium carbonate, and nutritional variables) based on Vis-NIR spectroscopy data. Compared with the existing works, the contributions of this study were summarized as follows.

(1) We proposed PS-MTL-LUCAS, a multi-task learning model to predict multiple soil properties on the public LUCAS topsoil dataset. PS-MTL-LUCAS adopted the partially shared layer to learn general informative representations and interact with each task-specific layer. This enabled to reduce the communication cost among layers during training.
(2) Feature wavelengths of each predictor were extracted by the SHapley Addictive exPlanations (SHAP) algorithm. The spectral responses of each soil property during prediction were revealed, enabling to reduce the dimensionality of the input data, while maintaining the prediction performance.
(3) Extensive evaluations were conducted to validate the effect of spectral pre-processing methods and the partially shared layer. The performance of PS-MTL-LUCAS and state-of-the-art models was compared. Lastly, the robustness of the extracted feature wavelengths was verified.

2. Materials and methods

2.1. LUCAS topsoil dataset

In this study, the Soil data of Land Use/Cover Area frame statistical Survey Soil (LUCAS topsoil) dataset in 2009 was used. 19,035 observations and their corresponding spectral files were accessible from this dataset (Orgiazzi et al., 2017; Tóth et al., 2013). The spectra ranged from 400.0 nm to 2499.5 nm at a resolution of 0.5 nm (4200 wavelengths in total for a single record).

In this study, we selected 7 indicators, including pH, N, P, K, cation exchange capacity (CEC), organic carbon (OC) and, $CaCO_3$, as the prediction targets. In general, these seven soil properties are key indicators of soil fertility (i.e., the ability of soil to provide plants with essential nutrients and favorable chemical, physical, and biological characteristics). For instance, CEC suggests the ability of soil to supply plant nutrients like calcium, magnesium, and potassium. Meanwhile, measuring the abundance of N, P, and K is critical, since these nutrients directly determined whether the soil is fertile for plant growth. Also, OC is important and it represents the amount of carbon retained in the soil after the decomposition of the organic content. The soil fertility drops as the organic content decreases.

We intended to have the complete dataset for training and validation. Therefore, we excluded the soil texture properties (e.g., sand, clay, and silt), since these properties did not have values for organic samples. The statistical summary of all 7 soil properties was shown in Table 1. It can be seen that except pH, the value range of each soil property was relatively large, and the data of P and K were skewed. This was consistent with the findings of previous studies (Tsakiridis et al., 2020). In Table 1, the indicator of skewness measured the asymmetry of a data distribution. It was noted that the skewness of soil properties, P and K, was high, reaching 6.68 and 8.93, respectively, which would be harmful during model training (Chen et al., 2022).

2.2. Spectral pre-processing

Before model training, we needed to perform spectral pre-processing operations on the raw spectral data (Ng et al., 2019; Padarian et al., 2019; Tziolas et al., 2020; Zhang et al., 2019). Previous studies have demonstrated that the performance of deep learning models depends on the spectral processing methods. It was worth mentioning that the convolutional kernels might also be treated as a spectral pre-processing method (Ng et al., 2019). In this study, the spectral profile ranging from 400.0 to 2499.5 nm was kept for calculation. We adopted the Savitzky-Golay (SG) filter following by Standard Normal Variable (SNV) as the pre-processing method. The zero-, first-, and second-order derivatives were applied to the SG filter, denoted by SG0, SG1, and SG2, respectively. The window size for the SG filter was set at 51 and the polynomial degree was set at 2. In brief, SG1 visualized the slope of the spectra, while SG2 could describe the noise distribution of the spectra. It was noted that only applying the SG filter without SNV would lead to a small value interval (e.g., from −0.002 to 0.012 for SG1 and from −0.0003 to 0.0001 for SG2), which was not ideal for further processing. Therefore, we applied the SNV operation after the SG filter to transform the spectra to a standard normal distribution.

Table 1
Statistics of the LUCAS topsoil dataset.

Soil properties	Valid samples	Min	Max	Mean	Standard deviation	Skewness
pH	19,035	3.21	10.08	6.20	1.35	−0.13
N/g•kg^{-1}	19,035	0	38.60	2.92	3.75	2.44
P/mg•kg^{-1}	19,035	0	1366.40	30.07	32.85	6.68
K/mg•kg^{-1}	19,035	0	7342.00	197.05	229.29	8.93
CEC/cmol (+)•kg^{-1}	19,035	0	234.00	15.76	14.48	1.94
OC/g•kg^{-1}	19,035	0	586.80	49.98	91.27	2.67
$CaCO_3$/g• kg^{-1}	19,035	0	944.00	51.60	125.31	2.87

Note: N, nitrogen; P, phosphorus; K, potassium; CEC, cation exchange capacity; OC, organic carbon; $CaCO_3$, Calcium carbonate.

2.3. PS-MTL-LUCAS modeling

2.3.1. Partially shared layer

A multi-task learning model aims at simultaneously learning multiple tasks through shared representations. The related tasks can take advantage of the complementary information to boost the overall performance by interaction among task-specific layers (Zhang and Yang, 2021). Generally, the hard parameter sharing and soft parameter sharing architectures were adopted when establishing MTL models (Thung and Wee, 2018), shown in Fig. 1(a) and 1(b), respectively. In the hard parameter sharing architecture, an MTL model learns a common representation for all tasks, meaning that all tasks share the weights completely. The MTL model with the hard parameter sharing architecture would be limited by the sensitivity of task weights since this set of weight parameters has to be generalized for all tasks (Nakamura et al., 2022). In the task of soil property prediction, it is noted that certain tasks may have potential conflicts. For instance, soil pH and K is poorly related during prediction (Khechba et al., 2021). This inherent conflict between tasks seriously reduces the prediction precision. In the soft parameter sharing architecture, constrained layers are introduced to regularize the distance between the parameters of the model, so that the parameters of related tasks would be similar. This approach provides a more flexible alternative to couple the shared representation sparsely (Yu and Xu, 2022). However, the scalability of the soft parameter sharing architecture remains as an open issue. Besides, sharing at high-level features are limited since there are no interactions among different tasks after the bifurcation.

To address the limitations of both the hard parameter sharing and soft parameter sharing architectures, we proposed a multi-task learning model with a partially shared structure inspired by Cao et al. (2018), shown in Fig. 1(c). Apart from preserving the task specific layers for each task, a partially shared net was introduced to the PS-MTL-LUCAS model. The partially shared net focused on learning and sharing the informative representations among tasks. Each partially shared layer took additional inputs from previous task specific layers. The concatenated feature map was then passed to the next partially shared layer and task specific layers, as shown in Fig. 2.

In Fig. 2, the information flow of each sub net and the partially shared net can be formulated by Eq. (1) and (2), respectively.

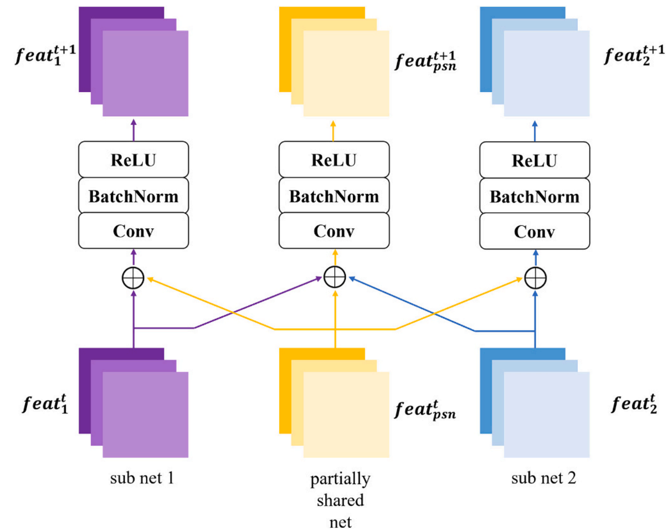

Fig. 2. The structure of the partially shared net.
Note: *feat* represents the feature map. *t* represents the serial number of layers. *Psn* is short for the partially shared net. For instance, $feat_{psn}^t$ denotes the feature map from the t^{th} layer of the partially shared net. Conv and BatchNorm represent the convolution and batch normalization operations, respectively. ReLU represents the Rectified Linear Unit activation function.

$$feat_1^{t+1} = F\left(c\left(feat_1^t, feat_{psn}^t\right)\right) \tag{1}$$

$$feat_{psn}^{t+1} = F\left(c\left(feat_1^t, feat_2^t, feat_{psn}^t\right)\right) \tag{2}$$

Where function $F(\cdot)$ indicates a sequence of operators (i.e., 1D convolution, batch normalization, and the ReLU activation function) and $c(\cdot)$ represents the concatenation operator.

In general, as one of the most essential components (or layers) in a deep neural network, an 1D convolution operator was used to automatically extract features from a given signal (e.g., the spectrum of a soil

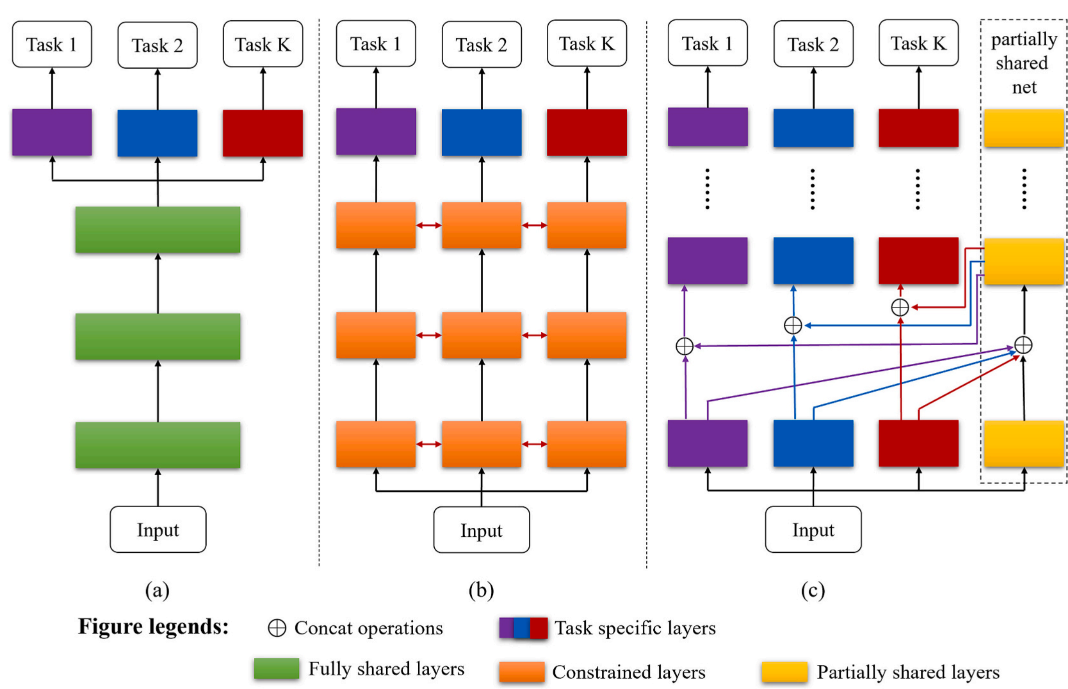

Fig. 1. Architectures of MTL models: (a) hard parameter sharing, (b) soft parameter sharing, (c) our proposal.

sample record). The convolution operator applied a kernel with a fixed size over the input and generated an output by the weighted average of all the inputs around it. Next, batch normalization (BatchNorm) was adopted to normalize the layer's inputs, thereby accelerating the convergence of the neural network. The BatchNorm operation first determined the mean and the variance of the activation values across the batch. Then a linear transformation with trainable parameters was applied to calculate the layer's output. It was widely acknowledged that the BatchNorm operation was especially helpful to tackle the gradient dispersion during model training. The ReLU activation function was typically employed to introduce non-linearity, assisting the neural network in learning the complex patterns from the input data. Last, the concatenation operator joined two feature maps by the channel dimension. This transformation was heavily adopted to fuse the features from different branches, which enriched the feature representation greatly.

2.3.2. Loss function

As we mentioned earlier, correlation existed in the soil property prediction tasks. According to Zhong et al. (2021), soil properties, such as N and OC, achieved a coefficient of determination (R^2) at 0.92, while pH and $CaCO_3$ reached R^2 at 0.52. Therefore, we further took advantage of these correlations to improve the performance of PS-MTL-LUCAS. In this study, we adopted the local constraint loss for the partially shared net (Schroff et al., 2015), formulated by Eq. (3).

$$Loss_{psn} = \frac{1}{N(N-1)} \sum_{i=1}^{N} \sum_{j=i+1}^{N} w_{i,j} \left\| feat_i^{t+1} - feat_j^{t+1} \right\|^2 \tag{3}$$

where N denotes the number of features and $w_{i,j}$ denotes the weight parameter. In particular, if samples i and j are similar, $w_{i,j}$ equals to 1, otherwise, it equals to 0.

In Eq. (3), the loss function of the partially shared net implied that tasks with close correlations may share similar attribute representations. By setting the local learning constraint, $Loss_{psn}$ prompted the attribute representation of related tasks to be close to each other, thereby enhancing the performance of PS-MTL-LUCAS.

Consequently, the total loss function of the PS-MTL-LUCAS can be formulated by Eq. (4).

$$totalLoss = \sum_{k=1}^{7} Loss_{sn\_i} + \alpha Loss_{psn} \tag{4}$$

where $Loss_{sn\_i}$ denotes the loss functions for 7 sub nets and we adopted the mean square error (MSE) here. α denotes the regularization parameter of $Loss_{psn}$, which controls the effect of the partially shared net. In this study, α was set at 0.3.

2.3.3. Sub net of PS-MTL-LUCAS

The structure of each sub net was established based on ResNet (He et al., 2016), shown in Fig. 3. ResNet was the champion of ImageNet Large Scale Visual Recognition Challenge (ILSVRC) 2015 and showed outstanding performance on the tasks like classification, regression, segmentation, detection, and so on. Although the model performance can be improved by stacking deeper layers, the gradient vanishing issue would occur during model training, leading to the fact that the deep neural network was degraded to shallow ones. To tackle this degradation issue, ResNet employed the residual building block by identify mapping, also known as the skip connection. This identify mapping allowed the input from the previous layer to be transferred directly to the deep layer, enabling to train much deeper networks.

In Fig. 3(a), each sub net was composed of convolutional layers, max pooling layers, residual blocks, and fully connected layers. As the major building block of PS-MTL-LUCAS, the convolutional layer acted as a filter and generated a feature map which activated the presence of detected features. In this study, we used the 1D convolution operator with a kernel size of 7. Next, the 1D max pooling operator with a kernel size of 3 calculated the maximum value in a given patch of the feature map. After this down-sampling operation, the dimensionality of the feature map was decreased. In regard to the fully connected layer, the neurons in this layer applied the linear transformation to the flattened vector by a weight matrix and then added a bias vector. Since predicting the soil property was a regression task, the shape of the output was 1. The neurons of two fully connected layers were set to 200 and 100, respectively, with a dropout rate of 0.3. The dropout unit would inactivate neurons randomly during training to assist PS-MTL-LUCAS in overcoming the overfitting issue.

Fig. 3(b) illustrated the internal composition of the residual block. The use of the residual block can avoid the gradient vanishing problem when the depth of the network increased (Alexeev et al., 2019; Gao

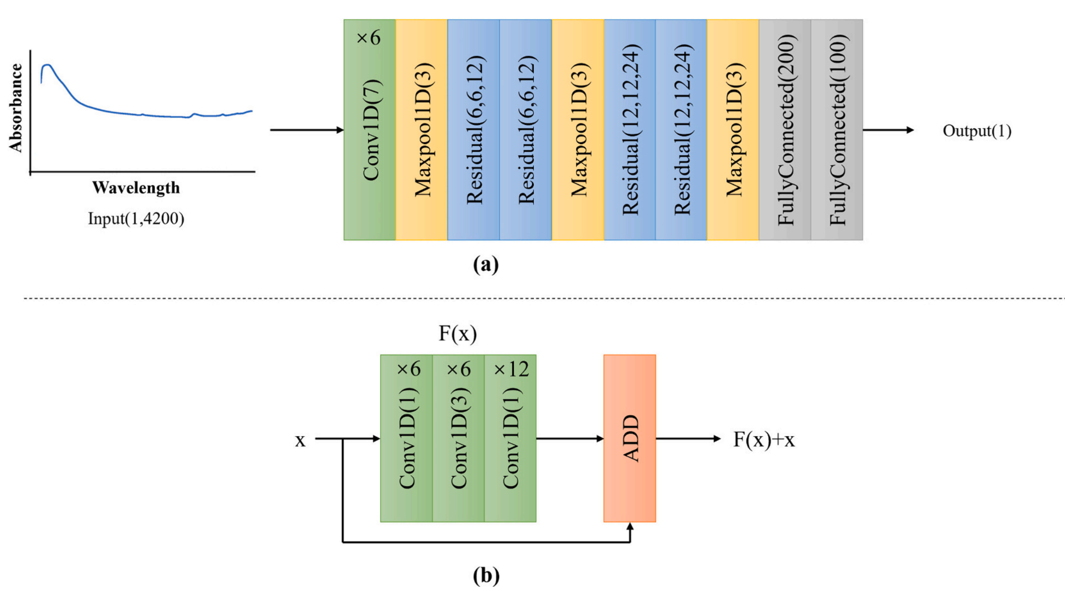

Fig. 3. The structure of each sub net: (a) diagram of the sub net, (b) diagram of the residual block.
Note: The input sample was randomly selected from the dataset. '×6' and '×12' denote the number of 1D convolution operators.

et al., 2021). In this study, the residual block consisted of stacked 1D convolutional layers with various window sizes and an ADD operator. Meanwhile, the input information can be passed to the subsequent layers through the skip connection, and eventually added to the feature map that generated by the normal weight layers.

In the PS-MTL-LUCAS model, the structure of the partially shared net remained the same as the sub net. The feature information exchanged after each max pooling layer. The feature map of each sub net was passed to the partially shared net, and the sub net treated the shared information as the input to the upcoming residual block or fully connected layers. The implementation of PS-MTL-LUCAS can be found at the following GitHub link (https://github.com/ZhaoyuZHAI/PS-MTL-LUCAS).

2.3.4. Spectral feature wavelength extraction

The raw Vis-NIR spectroscopy data was usually high-dimensional and contained redundant information. Therefore, applying feature extraction algorithms to identify the feature wavelength became a necessary procedure. Popular wavelength extraction algorithms included SHAP (Zhong et al., 2021), random forest (RF) (Cao et al., 2022), principal component analysis (PCA) (Deepa and Thilagavathi, 2015), and partial least square regression (PLSR) (Wu et al., 2014).

In this study, we chose the SHAP algorithm to extract the feature wavelengths by determining the positive or negative contributions of each wavelength to the given prediction. The calculation of the SHAP value followed previous publications (Haghi et al., 2021; Padarian et al., 2020), and it can be formulized by Eq. (5).

$$g(z') = \phi_0 + \sum_{i=1}^{4200} \phi_i z'_i \tag{5}$$

where g denotes the approximation of the original model, ϕ_0 denotes the base parameter, ϕ_i denotes the contribution of the i^{th} feature of the input, and z'_i indicates 1 when the i^{th} feature was observed, otherwise 0.

By calculating the SHAP value of each wavelength, their relative contributions can be evaluated. The spectral characteristic wavelength can be extracted according to the SHAP value ranking result.

For the RF algorithm, we evaluated the wavelength importance by the change of Out-of-Bag error after wavelength replacement. The higher the wavelength importance, the greater the variation in the prediction error rate would achieve. Regarding the PCA algorithm, it was a common data compression approach to reduce the data dimensionality. PCA mapped the features from the original space into a latent space and the retained k-dimensional features were viewed as principal components (PCs). Considering the PLSR algorithm, it extracted the most contributing wavelengths according to the regression coefficients resulting from the PLSR calibration model. It was worth noting that RF, PCA, and PLSR could be treated as self-interpretable feature wavelength extraction algorithms, because they did not get involved in the decision-making of PS-MTL-LUCAS. On the contrary, SHAP was a post-hoc explanation approach by evaluating the average marginal contribution of a feature among all possible combinations to PS-MTL-LUCAS.

In this study, SHAP was used to extract the feature wavelengths. For each soil property, we extracted the top 30 wavelengths that contributed the most to the prediction. Three new datasets were established for evaluating the robustness of the extracted wavelengths: (i) Top1-Top10 feature wavelengths for each soil property (70 wavelengths in total), (ii) Top1-Top20 feature wavelengths for each soil property (140 wavelengths in total), and (iii) Top1-Top30 feature wavelengths for each soil property (210 wavelengths in total).

2.3.5. State-of-the-art models for comparison

In this study, we selected five state-of-the-art models for comparison. First, the single-task learning models, DSNA (Piccoli et al., 2023), PCR-poly (Tavakoli et al., 2023), and ResNet-16 (Zhong et al., 2021) were chosen. For predicting each soil property, the single-task learning models were trained individually, meaning that there were 7 sets of weight parameters in total.

In regard to the multi-task learning model, Multi-LucasResNet-16 (Zhong et al., 2021) and Multi-gate Mixture-of-Experts (MMoE) (Ma et al., 2018) were selected for comparison. For the former one, it adopted the hard parameter sharing structure. The shared layer was developed based on the ResNet-16 and each soil property prediction task had its own fully connected layers with a dropout unit. While, the MMoE model employed the Mixture-of-Experts structure by sharing expert layers across all tasks. A gated mechanism was trained to optimize each task by allowing the useful information to be passed through.

2.3.6. Training details

The experiments were conducted using Pytorch 1.11.0 under the operation system Ubuntu 18.04 (64-bit), with Inter(R) Xeon(R) Platinum 8255C CPU, 64GB RAM. All models were established by Python 3.9.16 and trained by NVIDIA GeForce RTX 3090 (24GB) GPU. The LUCAS topsoil dataset was divided to a training dataset and a test dataset by the ratio of 4:1. The 5-fold cross-validation strategy was applied to guarantee the stability and robustness of all models. We used the Adam with Nesterov momentum algorithm as the optimizer. The learning rate was set at 0.0001, the batch size was set at 32, and the number of epochs was set at 1000.

2.3.7. Evaluation criteria

The coefficient of determination (R^2) and root mean square error (RMSE) were selected as the evaluation criteria. The formulation of these two evaluation criteria were defined by Eqs. (6) and (7). The larger the R^2 and the smaller the RMSE, the better the prediction performance and stability of the trained model would achieve.

$$R^2 = 1 - \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2 / \sum_{i=1}^{n} (y_i - \overline{y})^2 \tag{6}$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2} \tag{7}$$

where n denotes the number of samples, y_i denotes the observed value, $\widehat{y}_i$ denotes the predicted value, $\overline{y}$ denotes the mean of the observed values.

3. Results

3.1. Effect of spectral pre-processing

In this experiment, 6 spectral pre-processing algorithms were applied to the LUCAS topsoil dataset. The effects of each pre-processing algorithm on the PS-MTL-LUCAS model's prediction were shown in Fig. 4.

In Fig. 4, the effects of applying the SG0, SG1, and SG2 pre-processing algorithms had a major difference. For SG1 and SG2, the spectra would shift towards the center (zero value position), and multiple peaks appeared at around 500 nm, 1400 nm, and 1750 nm. To further explore the effect of these pre-processing algorithms, these pre-processed data, along with the raw data, were treated as inputs to the PS-MTL-LUCAS model. R^2 was used as the assessment indicator and the comparative result was shown in Table 2. The values presented in Table 2 were the average result of the 5-fold cross-validation on the test set.

From the result in Table 2, it can be seen that when inputting the raw Vis-NIR spectroscopy data to PS-MTL-LUCAS, the R^2 of predicting pH, N, OC, and $CaCO_3$ were all reached above 0.9, however, the PS-MTL-LUCAS model performed poorly when predicting the soil property P, achieving the R^2 at 0.378. On the whole, the pre-processing algorithms could boost the prediction ability of PS-MTL-LUCAS, especially for soil properties like P, K, and CEC. It was worth noting that PS-MTL-LUCAS

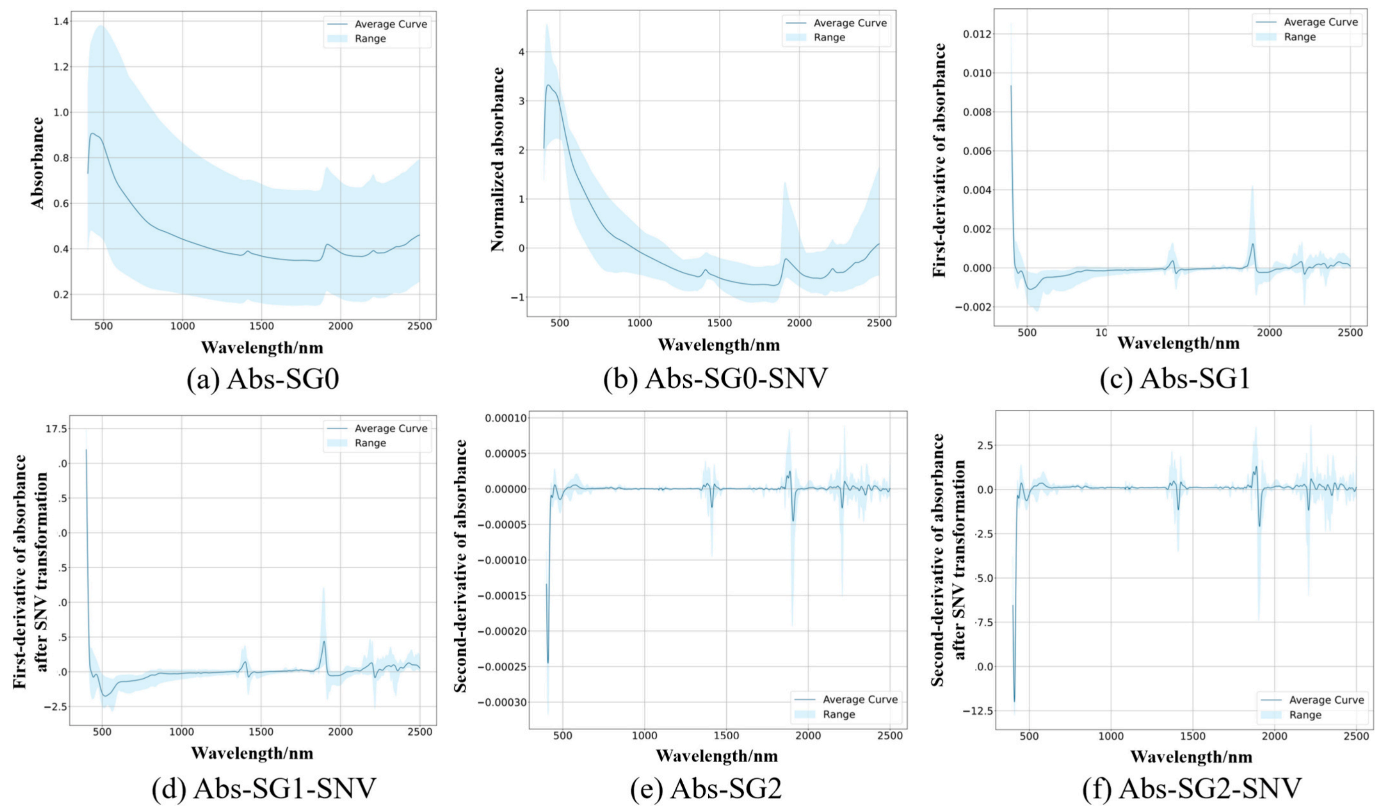

Fig. 4. The effect of 6 pre-processing algorithms: (a) SG0, (b) SG1, (c) SG2, (d) SG0-SNV, (e) SG1-SNV, (f) SG2-SNV.

Table 2
Comparative result of 6 pre-processing algorithms on PS-MTL-LUCAS.

Soil properties	Raw	SG0	SG1	SG2	SG0-SNV	SG1-SNV	SG2-SNV
pH	0.928	0.928	0.928	0.930	0.933	**0.945**	0.935
N	0.931	0.924	0.918	0.918	0.926	**0.936**	0.926
P	0.378	0.388	0.403	0.399	0.410	0.413	**0.415**
K	0.585	0.611	0.613	0.607	0.609	**0.624**	0.613
CEC	0.792	0.798	0.801	0.799	0.807	**0.837**	0.818
OC	**0.953**	0.930	0.929	0.929	0.939	0.952	0.943
$CaCO_3$	0.944	0.941	0.943	0.943	0.942	**0.956**	0.951

Note: The evaluation indicator in this table was R^2 on the test dataset. The bold font indicated the optimal value.

achieved the optimal performance of predicting N and OC with the raw Vis-NIR spectroscopy data. The reason about this performance drop after pre-processing could be that some spectral features might be eliminated through the smoothing operation. The experimental result also showed that the SNV operation could further improve the pre-processing effect. Among all 6 pre-processing algorithms, SG1-SNV prompted PS-MTL-LUCAS to achieve the optimal performance for predicting pH, N, P, K, CEC, and $CaCO_3$, while the precision of predicting OC ranked the second, right after inputting the raw data to the model. Therefore, we determined to adopt SG1-SNV as the pre-processing algorithm in the following experiments.

3.2. Comparison of different models

We compared PS-MTL-LUCAS with ResNet-16, Multi-LucasResNet-16, MMoE, PCR-poly, and DSNA in the task of predicting 7 soil

Table 3
Performance comparison of all models on the test dataset.

Soil properties	Assessment indicators	PS-MTL-LUCAS	Multi-LucasResNet-16	MMoE	ResNet-16	PCR-poly	DSNA
pH	R^2	**0.945**	0.942	0.823	0.930	0.940	0.903
	RMSE	**0.319**	0.324	1.485	0.357	0.350	0.431
N	R^2	**0.936**	0.932	0.831	0.925	0.920	0.792
	RMSE ($g \bullet kg^{-1}$)	**0.895**	0.977	1.984	1.052	1.110	0.737
P	R^2	**0.413**	0.398	0.357	0.352	NA	0.367
	RMSE ($mg \bullet kg^{-1}$)	**17.237**	20.847	24.156	24.743	NA	23.914
K	R^2	**0.624**	0.587	0.513	0.570	NA	0.544
	RMSE ($mg \bullet kg^{-1}$)	**109.671**	135.064	187.461	131.071	NA	167.836
CEC	R^2	**0.837**	0.789	0.725	0.733	0.800	0.731
	RMSE ($cmol(+) \bullet kg^{-1}$)	**6.018**	6.891	8.017	7.956	6.890	5.618
OC	R^2	**0.952**	0.950	0.929	0.942	0.950	0.830
	RMSE ($g \bullet kg^{-1}$)	**18.691**	19.983	25.285	22.347	21.340	21.7225
$CaCO_3$	R^2	0.956	0.957	0.929	0.952	**0.960**	0.937
	RMSE (g•kg-1)	26.673	25.676	31.468	27.691	**25.710**	32.502

Note: The bold font indicated the optimal value.

properties simultaneously. The comparative result was presented in Table 3. For the models like PS-MTL-LUCAS, Multi-LucasResNet-16, MMoE, and ResNet-16, they were all trained and validated on the same dataset split, while for PCR-poly and DSNA, the results were referred from the published articles.

The result from Table 3 suggested that the PS-MTL-LUCAS model dominated in almost all the soil property prediction tasks, apart from predicting $CaCO_3$, where a very tiny gap was detected when being compared with Multi-LucasResNet-16. PS-MTL-LUCAS performed particularly well in predicting pH, N, OC, and $CaCO_3$, achieving the R^2 at 0.945, 0.936, 0.952, and 0.956, respectively. The value of RMSE for these tasks indicated the same. Among all three multi-task learning models, MMoE achieved the worst performance. Based on the principle of MMoE, it was good at modeling task relationships by utilizing the shared information by the gating mechanism. However, the correlation between most soil properties was relatively low, leading to the fact that the information shared by each expert layer would be limited. Considering the single-task learning model, the PCR-poly model achieved the optimal performance, particularly, it can predict the $CaCO_3$ content well, reaching the highest R^2 at 0.960. It was also noted that all 6 models failed to predict the soil property of P accurately. The highest R^2 and the lowest RMSE were obtained by PS-MTL-LUCAS, at 0.413 and 17.237 mg•kg^{-1}, respectively. The other three models achieved the R^2 all below 0.4.

The scatter plots of values predicted by PS-MTL-LUCAS and the ground truth for 7 soil properties were provided in Fig. 5. We used the seaborn library (a Python data visualization library based on matplotlib) to draw Fig. 5 and visualized the scatter point density. It can be intuitively seen from the fitting line that PS-MTL-LUCAS was able to generate accurate prediction results.

3.3. Feature wavelength extraction results

Taking the feature wavelengths of pH as an example, the SHAP values of Top 1-Top10 wavelength were shown in Fig. 6. It can be concluded that the feature wavelengths of pH were concentrated around 1700 and 2010 nm, which contributed the most to predict the soil pH. The top 3 contributing wavelengths were 1700.5, 2008.5, and 2010.0 nm, respectively.

Also, we calculated the SHAP values of each wavelength and visualized the importance of all wavelengths for predicting each soil property through a heatmap. The result was presented in Fig. 7.

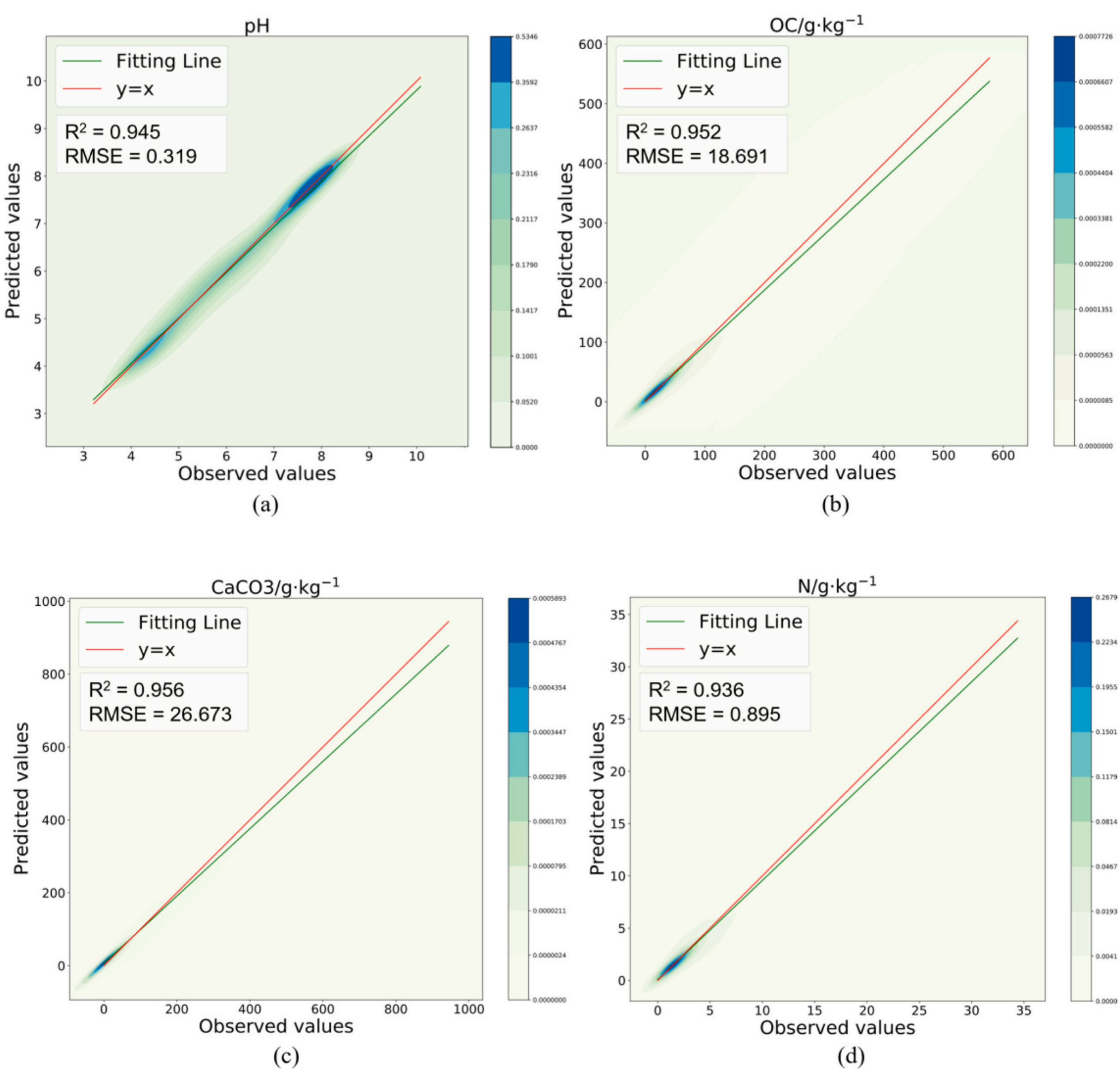

Fig. 5. The scatter plots of predicted values and ground truth for PS-MTL-LUCAS: (a) pH, (b) OC, (c) $CaCO_3$, (d) N, (e) P, (f) K, and (g) CEC.

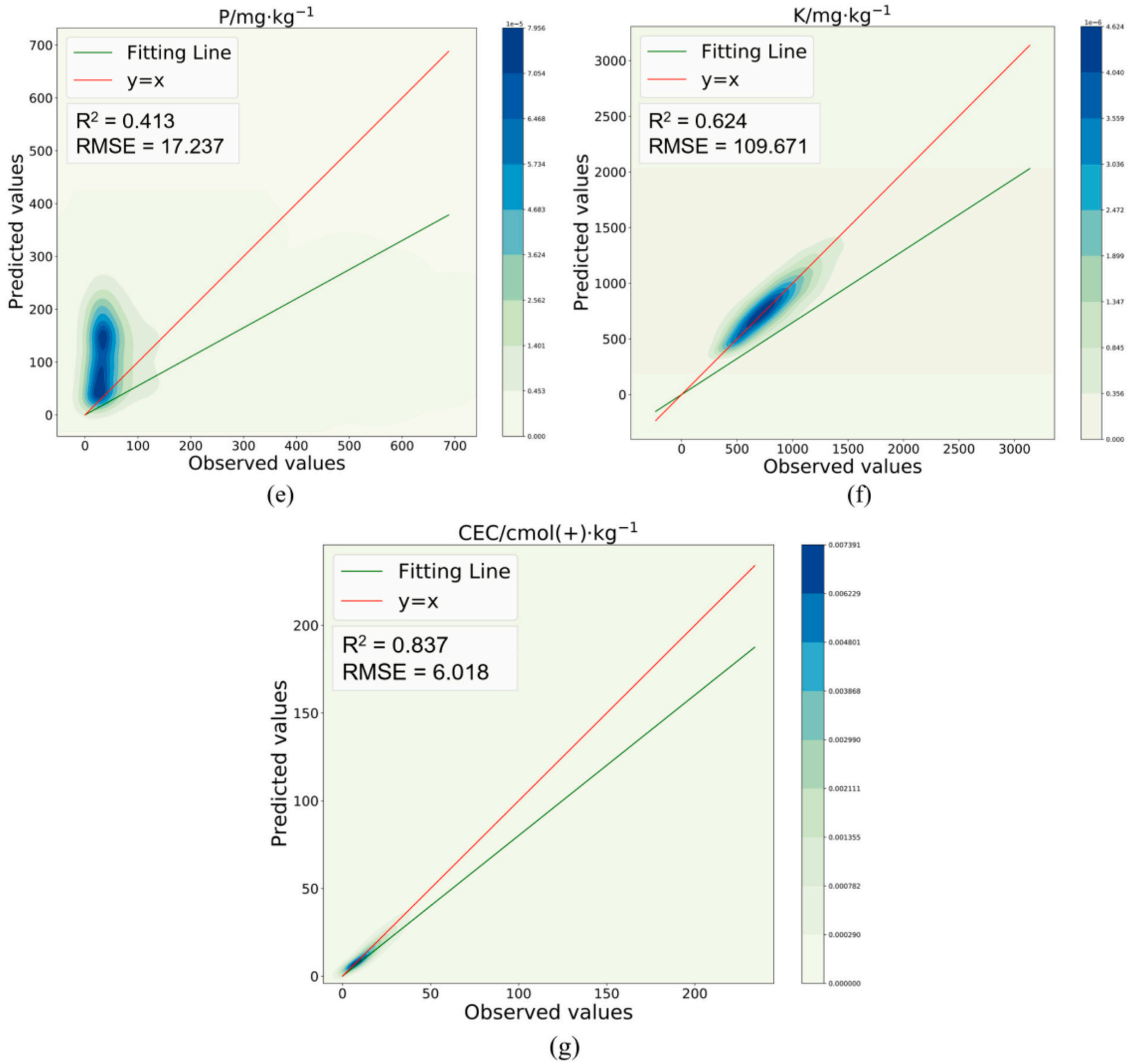

Fig. 5. (*continued*).

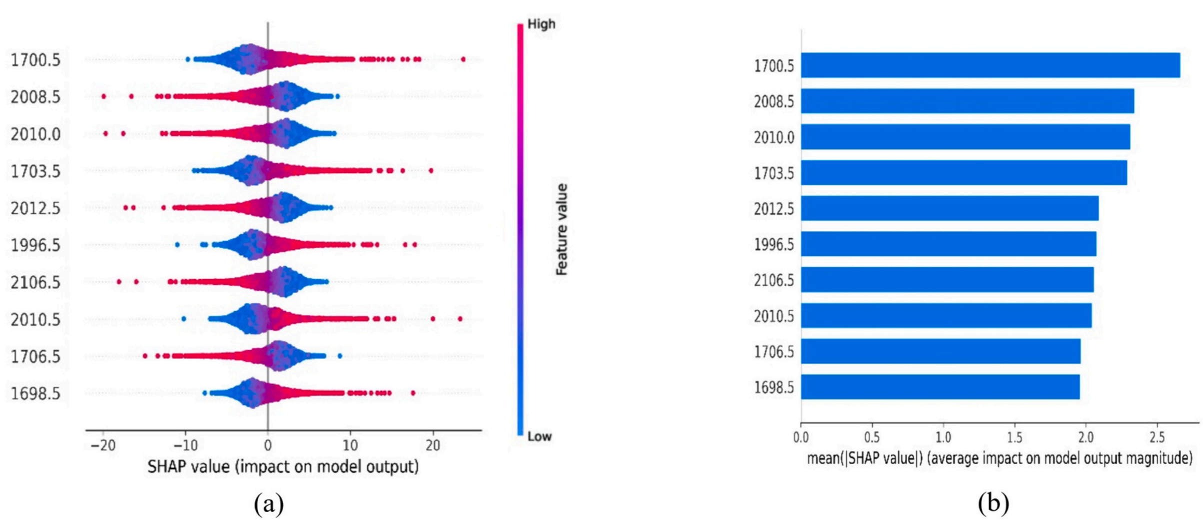

Fig. 6. SHAP values of feature wavelengths for pH: (a) the bee swarm plot, and (b) the bar plot.

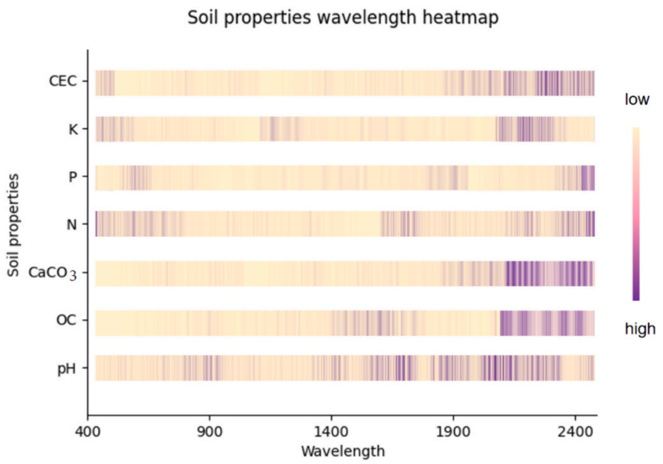

Fig. 7. Contributions of all wavelengths to each soil property.

For verifying the robustness of the extracted feature wavelengths, we also conducted a comparative experiment between different extraction algorithms. SHAP was compared with 3 algorithms, including PLSR, PCA, and RF. Each algorithm was employed to extract the Top10 feature wavelengths and treated these wavelengths as inputs (70 wavelengths in total) to the PS-MTL-LUCAS model. Still, the 5-fold cross validation strategy was applied and the training dataset and test dataset were divided by the ratio of 4:1. R^2 was used as the assessment indicator. The comparative result was shown in Table 5, suggesting that the feature wavelengths extracted by SHAP were more effective to predict all soil properties. Meanwhile, we noted that the task of predicting soil properties of P and K encountered a dramatically drop.

3.4. Effect of input numbers on the model performance

We further tested whether different number of inputs might have an impact on the PS-MTL-LUCAS model. We selected the 70, 140, 210 feature wavelengths (i.e., 10, 20, and 30 feature wavelengths for each soil property respectively) extracted by SHAP as inputs, and compared them with inputting the full wavelengths (4200 wavelengths in total). The value of R^2 was shown by a bar chart in Fig. 8.

From the result in Fig. 8, we can see that when inputting the total 4200 data points, PS-MTL-LUCAS achieved the optimal performance in all prediction tasks. When inputting the top1 to top 30 wavelengths of each soil property to PS-MTL-LUCAS, the prediction performance experienced an average decrease around 6% for all soil properties. When excluding the soil properties K and P, the average decrease was around 3%. The prediction precision dropped as the number of input data points decreased. The worst performance was achieved by the one with 70 data points. This was reasonable because the extracted feature wavelengths contained limited information and they were unable to represent the soil samples well. In summary, except for the soil properties P and K, the prediction results of other properties were acceptable.

4. Discussions

4.1. Multi-task learning or single-task learning?

From the result in Table 3, it can be concluded that the multi-task learning models generally could achieve better performance than single-task learning. Although single-task learning achieved an acceptable result in most prediction tasks, massive training resources were required since each prediction task had to be trained individually. For predicting all 7 soil properties, 7 sets of weight parameters were required (Xu et al., 2018). Therefore, it would be advantageous to build all single-task learning models simultaneously using the multi-task learning structure (Qi et al., 2017). Because the generalization of the prediction model can be significantly improved by making use of the shared information among related tasks (Ge et al., 2019; Xu et al., 2017). Besides, multi-task learning models had the advantage of scalable and modular framework (He et al., 2019). In case that 1 or 2 soil properties were missing, we can easily remove the corresponding sub nets (i.e., the task-specific layers) and proceed the training process without any further changes. The scalability of multi-task learning models also enabled to train with more soil properties by simply adding more sub nets.

Among the multi-task learning models, one exception appeared for the MMoE model (see Table 3). MMoE performer slightly worse than ResNet-16. Theoretically, MMoE was good at modeling task relationships (Wang et al., 2022). The performance of MoE layers extremely depended on the relatedness among tasks. When the correlation

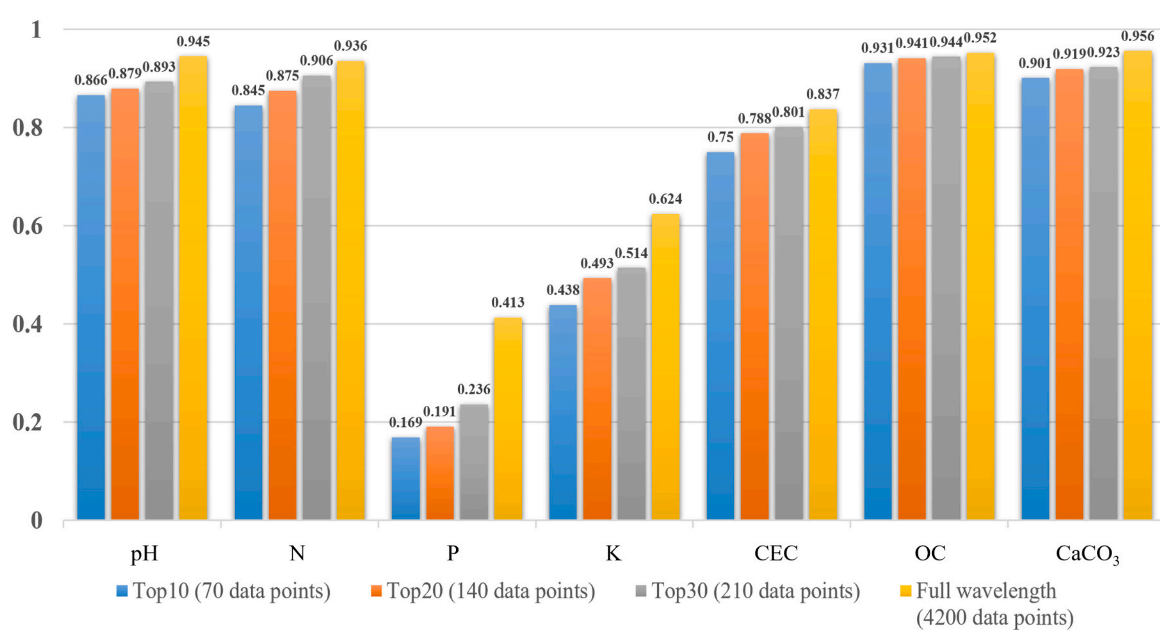

Fig. 8. Comparative result of different numbers of inputs.

between two tasks reached 0.9 (i.e., almost 2 identical tasks), the MoE layer would work better, but if the correlation value dropped to 0.5, the performance would decrease accordingly (Ma et al., 2018). In fact, the correlation between soil properties was weak in the LUCAS topsoil dataset (Zhong et al., 2021). Only 3 pairs (OC and N, CEC and N, pH and $CaCO_3$) achieved a correlation above 0.5. Consequently, the MoE layer might not take effect. On the contrary to the MMoE model, PS-MTL-LUCAS utilized a partially shared net during training. After feature maps generated by the max pooling layer in each sub net, the information was exchanged within the partially shared net and the joint knowledge was passed to each sub net in the next layer to improve the prediction performance.

4.2. Balance between the prediction performance and the number of inputs

Considering the deployment of PS-MTL-LUCAS in real world applications, it is necessary to balance the prediction performance and the required data points used for training. Acquiring Vis-NIR spectroscopy data is costly due to its high dimensionality (Shen et al., 2020; Tian et al., 2022). The LUCAS topsoil dataset used in this study recorded the full continuous spectrum ranging from 400 to 2500 nm with a spectral resolution of 0.5 nm, resulting in 4200 data points in a single record. These data points were the output of the instrument and they did not represent the true spectral resolution. Previous studies might perform a resampling operation before feeding to the deep learning models (Qi et al., 2017). Although the experimental result in this study suggested that PS-MTL-LUCAS would achieve the optimal performance when all the 4200 data points were input to the model, using extracted feature wavelengths to predict multiple soil properties was also feasible. By reducing the number of input data points, the expense of data acquisition can be greatly reduced and the computational efficiency would be enhanced (Cao et al., 2021; Zhuang et al., 2020). Also, the dimensionality reduction algorithms enabled to eliminate the redundant and noisy information, thereby improving the applicability of PS-MTL-LUCAS. We noticed that various feature wavelength extraction methods had been used in previous studies, such as the elastic net (He et al., 2023), grey wolf optimizer algorithm (Tian et al., 2022), successive projections algorithm (Jia et al., 2017), and so on. These methods may have different effects on the prediction performance. For the LUCAS topsoil dataset, the suitable one was SHAP, according to the result in Table 5. SHAP showed dominating performance over the other three (i.e., PLSR, PCA, and RF) and the wavelength extracted by SHAP had better interpretability.

According to the result from Fig. 7, the feature wavelengths for each soil property can be briefly summarized as follows: (i) for CEC, the feature wavelengths were around 597 nm and 2420 nm, (ii) for K, around 660 nm and 2230 nm, (iii) for P, around 467 nm, 579 nm, and 2310 nm, (iv) for N, around 410 nm, 2210 nm, and 2220 nm, (v) for $CaCO_3$, around from 2340 nm, (vi) for OC, around from 2240 nm, and lastly (vii) for pH, around 1700 nm and 2010 nm. For a more detailed look, we listed the Top10 feature wavelengths of each soil property in Table 4. The identification of these feature wavelengths was consistent with previous publications. For instance, Wang et al. (2021) screened out the characteristic wavelength by a recursive feature elimination and found that the wavelength at 2065 nm and 2071 nm were particularly helpful when estimating the soil total nitrogen content. These two wavelengths were close to our result (i.e., 2061 nm). Zhong et al. (2021) discovered that the wavelength ranging from 2180 to 2359 nm contributed the most to predicting the soil property of OC. Our finding also supported this result.

The performance of predicting soil properties P and K was poor for all models, no matter inputting all the 4200 data points or feature wavelengths. A potential explanation could be that the soil properties P and K had a narrow chemical range. Meanwhile, the wide data distribution of P and K might also be a disturbance factor for model calibration. This finding was consistent with previous publications. Ji et al. (2014) used visible and near-infrared spectroscopy (ranging from 350 to 2500 nm) to measure soil properties in paddy soil and the R^2 of predicting P and K only reached 0.43 and 0.14, respectively. Similar results can also be found in Schirrmann et al. (2013), Qi et al. (2017), and Zhong et al. (2021).

4.3. Limitations of PS-MTL-LUCAS

Through extensive experiments, the effectiveness of PS-MTL-LUCAS had been verified. However, in order to further improve the prediction performance of PS-MTL-LUCAS and deploy it in the real-world application, the following limitations should be addressed.

First, the feature maps generated by the pooling layer of each sub net and the partially shared net were now merged via the concatenate operation (Yu and Xu, 2022). It is considerable whether a weighted shared layer could be integrated to the PS-MTL-LUCAS model. For instance, Lopes et al. (2023) designed a cross-task attention mechanism for dense multi-task learning, which exploited pair-wise cross-task exchange through correlation guided attention and self-attention. Liu et al. (2019) proposed a multi-task attention network by utilizing a global feature pool, together with a soft-attention module for each task. Motivated by these works, we assumed that the PS-MTL-LUCAS model could be further enhanced by introducing an attention mechanism to highlight important features from each sub net and to determine their contribution to reduce the loss of the partially shared net.

Second, it is necessary to segment the soil background from the raw images. It has been widely acknowledged that the spectral responses of

Table 5
Comparative result of 4 algorithms for extracting feature wavelengths.

Soil properties	SHAP	PLSR	PCA	RF
pH	**0.866**	0.845	0.839	0.716
N	**0.845**	0.807	0.825	0.757
P	**0.169**	0.099	0.168	0.118
K	**0.438**	0.378	0.414	0.329
CEC	**0.750**	0.679	0.630	0.638
OC	**0.931**	0.861	0.879	0.861
$CaCO_3$	**0.901**	0.825	0.843	0.896

Note: The evaluation indicator in this table was R^2 on the test dataset. The bold font indicated the optimal value.

Table 4
The Top10 feature wavelengths of each soil property.

Soil properties	Top1	Top2	Top3	Top4	Top5	Top6	Top7	Top8	Top9	Top10
pH	1700.5	2008.5	2010.0	1703.5	2012.5	1996.5	2106.5	2010.5	1706.5	1698.5
N	2215.5	2221.0	2219.5	2219.0	407.0	2217.0	2357.0	413.5	2061.0	415.0
P	2313.5	1913.5	2309.0	2315.5	467.0	1917.0	579.0	2307.5	2305.5	2353.5
K	2229.0	665.0	657.0	661.0	2233.5	658.0	2236.0	666.0	1418.0	1916.0
CEC	593.0	597.0	2419.0	577.0	597.5	513.0	573.0	1427.0	598.5	2420.5
OC	2222.0	2244.0	2224.0	2245.5	2248.0	2324.0	2309.5	2246.0	2268.0	2241.5
$CaCO_3$	2345.5	2346.5	2344.5	2347.5	2343.5	2342.5	2350.0	1373.5	2338.0	2341.5

Note: The unit of all data is nm.

plants and soil have a major difference. In order to accurately predict the soil properties, we should only focus on the pixels that belong to soil (Hamuda et al., 2016; Zamani et al., 2022). At current, we just completed the soil property prediction task since the LUCAS topsoil dataset had already provided the data sources ready for use. However, when conducting field experiments, we have to perform the segmentation task in advance. State-of-the-art segmentation algorithms like 3DUnetGSFormer (Jamali et al., 2022), DeepLab (Chen et al., 2018), SegNet (Badrinarayanan et al., 2017), and so on, would be our potential choices in the upcoming research.

Last but not the least, incorporating the soil knowledge into the prediction of soil properties would be beneficial. Ma et al. (2023) encouraged that apart from pursuing the smallest root mean square error or largest R^2 values, soil scientists and computer algorithm developers should work together to comprehensively understand the soil itself, leading to better informed interpretations of the model's predictions. Integrating soil knowledge like soil genesis, soil classification, mineral assemblages into the modeling process could enhance the prediction accuracy of deep learning models. Furthermore, Weindorf and Chakraborty (2024) argued that although the application of deep learning models in soil science had achieved great success, soil knowledge from soil scientists' experience still remained crucial. Soil science encompassed complex interactions between multiple factors, including chemical, physical, and biological components. Soil scientists were also responsible to validate and interpret the results generated by deep learning models. Therefore, our next work on PS-MTL-LUCAS was to collaborate the deep learning models and human perspective to comprehensively understand the correlation between soil systems and Vis-NIR spectroscopy data.

5. Conclusion

By taking advantage of multi-task learning, PS-MTL-LUCAS was proposed to predict multiple soil properties, including OC, N, P, K, CEC, pH, and $CaCO_3$ content. The main contribution of this study was to propose a multi-task learning model with a partially shared layer. By interacting with the partially shared net, common knowledge was transferred between each sub net (i.e., task-specific layer), thereby enhancing the prediction performance of all tasks. For extracting the feature wavelength of each soil property, SHAP was adopted to evaluate the contribution of all wavelengths. By comparing with state-of-the-art models, the experimental results proved the effectiveness of PS-MTL-LUCAS and it showed dominating performance of predicting all soil properties, except for $CaCO_3$ (with a very tiny drop being compared with Multi-LucasResNet-16). The extracted feature wavelengths were also validated by comparing with the ones obtained by RF, PCA, and PLSR. Among them, SHAP achieved the optimal performance.

This study provided a new and novel soil property prediction method for soil studies and illustrated the correlation between the spectral responses and soil properties. For future work, we would further improve the prediction performance by introducing the attention mechanism (e.g., cross-task attention) to generate dynamic weights for each feature map during information sharing. We also consider to verify the robustness of PS-MTL-LUCAS on more soil samples and deploy our model in practice. Due to the nature of the LUCAS topsoil dataset, PS-MTL-LUCAS is now only applicable in the laboratory. For transferring the PS-MTL-LUCAS model to field conditions, it may require a segmentation operation in advance for separating the soil target from the background. So, some SOTA segmentation algorithms like 3DUnetGSFormer, DeepLab, and SegNet could be adopted in the upstream procedure. Last but not the least, utilizing expert knowledge from soil scientist would be beneficial to further interpret the correlation of Vis-NIR spectroscopy data and soil properties.

CRediT authorship contribution statement

Zhaoyu Zhai: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Fuji Chen:** Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Hongfeng Yu:** Visualization, Software, Methodology, Formal analysis. **Jun Hu:** Formal analysis, Data curation. **Xinfei Zhou:** Writing – review & editing, Supervision, Conceptualization. **Huanliang Xu:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Data availability

Data will be made available on request.

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